

QINGLIN MA

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RESEARCH

- The interactions between the large-scale structure and halos/galaxies.
- The formation and evolution of the dark matter halo and its coevolution with galaxy.
- Connecting galaxies to dark matter halos using spectroscopic survey.

EDUCATION

Ph.D. in Astronomy, **Tsinghua University** *Sept. 2021 - July. 2027 (expected)*

Supervisors: Cheng Li. Thesis: Evolution of the dark matter halo and its impact on the galaxy formation

Visiting Scholar, **Leiden Observatory** *Nov. 2025 - July. 2026*

Supervisors: Joop Schaye.

B.S. in Astronomy, **Sun Yat-sen University** *Sept. 2017 - Jun. 2021*

Supervisors: Xiaodong Li.

HONORS

Tsinghua Scholarship for the Visiting Scholar *Nov. 2025 - Mar. 2026*

1st Comprehensive Scholarships *2025*

Excellent Oral presentation in the 26th Guo Shoujing Symposium *2024*

Excellent Graduate Student of Sun Yat-sen University *2021*

TEACHING

- Observational Cosmology. Teaching Assistant, Tsinghua University, Autumn 2022
- Machine learning in astronomy Teaching Assistant, Sun Yat-sen University, Spring 2017

SERVICE

- Co-leader of weekly Text club at Department of Astronomy in Tsinghua University *2022-2025*
- Volunteer of Conference "The hundred-billion-year evolution law of galaxy ecosystems" *Jul, 2025*
- Co-organizer of weekly Galaxy-Cosmology Seminar at Department of Astronomy in Tsinghua University *2023*
- Faculty Candidate Interview Committee at Department of Astronomy in Tsinghua University, Phd Representative *2023*

CONFERENCE TALKS

- European Astronomical Society Annual Meeting 2026 (**Contributed talk**) *Lausanne, 2026. Jul*
Three-stage assembly of massive galaxies in the COLIBRE simulation
- European Astronomical Society Annual Meeting 2026 (**Contributed talk**) *Lausanne, 2026. Jul*
Interpreting the strong clustering of ultra-diffuse galaxies by halo spin bias
- 12th Sino-German Meeting (**Lightning talk**) *Chengdu, 2025. Sep*
The coevolution between galaxies and dark matter halos in FIRE-2 Simulations
- The hundred-billion-year evolution law of galaxy ecosystems (**Lightning talk**) *Beijing, 2025. Jul*
The coevolution between galaxies and dark matter halos in FIRE-2 Simulations

- Expanding the boundaries of dark matter halo (**Contributed talk**) *Shanghai, 2025. Mar*
Dependence of dark matter halo assembly bias on halo definition and orientation
- PFS collaboration meeting (**Lightning talk**) *Tokyo, 2025. Jan*
Dependence of dark matter halo assembly bias on halo definition and orientation
- 2024 PKU International PhD Student Forum (**Contributed talk**) *Beijing, 2024. Dec*
Lopsided and Bulging Distribution of Satellites around Paired Central Galaxies
- The 26th Guo Shoujing Symposium (**Contributed talk**) *Nanjing, 2024. Mar*
Dependence of dark matter halo assembly bias on halo definition and orientation
- 2nd Shanghai Assembly on Cosmology and Structure Formation (**Contributed talk**) *Shanghai, 2023. Oct*
Dependence of dark matter halo assembly bias on halo definition and orientation

SEMINAR

- Seminar talk, Max Planck Institute for Astrophysics (**invited**) *Germany, 2026. Jun*
Galaxy Structural Evolution Across Mass Scales: From Dwarfs to Massive Galaxies
- Group talk in Annalisa Pillepich' group, Max Planck Institute for Astronomy (**invited**) *Germany, 2026. Jun*
Galaxy Structural Evolution Across Mass Scales: From Dwarfs to Massive Galaxies
- Lunch talk, Leiden Observatory (**invited**) *Netherlands, 2026. Jan*
Interpreting the strong clustering of ultra-diffuse galaxies by halo spin bias
- Lightning talk, Department of Astronomy in Tsinghua University *Chengdu, 2025. Sep*
The coevolution between galaxies and dark matter halos in FIRE-2 Simulations

PUBLICATIONS

- ♦ 11 publications; 7 as the first/corresponding author (4 published, 1 submitted, 2 to be submitted); open in [NASA/ADS](#)

First and Corresponding[†] author papers:

1. **Qinglin Ma[†]**, Yangyao Chen, Houjun Mo & Joop Schaye, to be submitted in **MNRAS**
[Brightest Cluster Galaxies in the COLIBRE Simulation: A Three-Stage Assembly Pattern](#)
2. **Qinglin Ma[†]**, Yangyao Chen & Houjun Mo, to be submitted in **MNRAS**
[A fly-by origin of halo secondary bias](#)
3. **Qinglin Ma** & Cheng Li[†], submitted to **APJ** (2026) [ArXiv: 2607.26009]
[Anisotropic Secondary Bias of Dark Matter Haloes in a \$\Lambda\$ CDM Universe](#)
4. **Qinglin Ma**, Cheng Li[†], Yangyao Chen & Houjun Mo, accepted by **MNRAS** (2026) [ArXiv: 2512.17742]
[Interpreting the strong clustering of ultra-diffuse galaxies by halo spin bias](#)
5. **Qinglin Ma**, Yangyao Chen[†] & Houjun Mo, **MNRAS** 548 3 (2026) [ArXiv: 2509.16138]
[Two-phase formation of galaxies: the coevolution between galaxies and dark matter haloes](#)
6. **Qinglin Ma**, Cheng Li[†] & Yanhan Guo, **APJ** 922 1 (2025) [ArXiv: 2502.14668]
[Lopsided and Bulging Distribution of Satellites around Paired Halos. II. 3D Analysis and Dependence on Projection and Selection Effects](#)
7. **Qinglin Ma**, Guo, Yiqing, Li, Xiao-Dong[†], Wang, Xin, Miao, Haitao, Li, Zhigang, Sabiu, Cristiano G & Park, Hyunbae, **APJ** 890 2 (2020) [ArXiv: 1908.10595]
[Cosmological Constraints from the Redshift Dependence of the Alcock-Paczynski Effect: Possibility of Estimating the Nonlinear Systematics Using Fast Simulations](#)

Second-author publications:

1. Yanhan Guo, **Qinglin Ma** & Cheng Li, **APJ** 922 1 (2025) [ArXiv: 2502.14666]
[Lopsided and Bulging Distribution of Satellites Around Paired Halos. I. Observational Measurements and Comparison with Halo-based Models](#)

Co-author papers:

1. Ce Gao, Cheng Li, Houjun Mo, Jiacheng Meng, **Qinglin Ma**, Xiaohu Yang, Yizhou Gu, Qingyang Li, accepted by **MNRAS** (2026) [ArXiv: 2604.04794]
[Galaxy Populations in Groups and Clusters: II. Conditional Luminosity Functions at Redshifts from \$z \sim 1\$ to \$z \sim 0\$](#)
2. Shuren Zhou, Zhenjie Liu, **Qinglin Ma**, Yu Liu, Le Zhang, Xiao-Dong Li, Yang Wang, Xin Wang, Yu Yu, Hao-Ran Yu & Yi Zheng, **MNRAS** 512 3 (2021) [ArXiv: 2108.12568]
[Sensitivity tests of cosmic velocity fields to massive neutrinos](#)
3. Yizhao Yang, Haitao Miao, **Qinglin Ma**, Miaoxin Liu, Cristiano G Sabiu., Jaime Forero-Romero, Yuanzhu Huang, Limin Lai, Qiyue Qian, Qiyue Zheng & Xiao-Dong Li, **APJ** 900 1 (2021) [ArXiv: 2007.03150]
[Using the Mark Weighted Correlation Functions to Improve the Constraints on Cosmological Parameters](#)

REFERENCES

Prof. Cheng Li
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Tsinghua university

Prof. Houjun Mo
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University of Massachusetts, Amherst

Prof. Joop Schaye
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Leiden Observatory